# Algorithm Design: Homework 1

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## Problem 5

*(Source: KT book pp. 108, Problem 6.)*

We have a connected graph $G = (V, E)$, and a specific vertex $u \in V$. Suppose we compute a depth-first search tree rooted at $u$, and obtain a tree $T$ that includes all nodes of $G$. Suppose we then compute a breadth-first search tree rooted at $u$, and obtain the same tree $T$. Prove that $G = T$. In other words, if $T$ is both a depth-first search tree and a breadth-first search tree rooted at $u$, then $G$ cannot contain any edges that do not belong to $T$.

**Answer:**

Since $T$ is a DFS tree, every non-tree edge of $G$ joins a node to one of its non-parent ancestors in $T$.

Since $T$ is also a BFS tree, every edge of $G$ joins two nodes whose depths in $T$ differ by at most 1.

Suppose for contradiction that $G$ contains a non-tree edge. By the previous paragraph, one endpoint is a non-parent ancestor of the other, so their depths in $T$ differ by at least 2, contradicting the bound above.

Therefore no non-tree edge exists in $G$, which implies $G = T$.

## Problem 6

We are given $2n$ balls and $n$ bins. Each ball is thrown into a random bin (each bin is chosen with probability $1/n$). Let $X_n$ denote the number of empty bins after all balls have been thrown. Compute the following limit:

$$\lim_{n \to +\infty} \mathbb{E}\left[\frac{X_n}{n}\right].$$

**Answer:**

By the linearity of expectation, we have

$$\mathbb{E}\left[\frac{X_n}{n}\right] = \frac{1}{n} \cdot n \cdot \left(\frac{n-1}{n}\right)^{2n},$$

$$\lim_{n \to +\infty} \mathbb{E}\left[\frac{X_n}{n}\right] = \lim_{n \to +\infty} \left(1 - \frac{1}{n}\right)^{2n} = e^{-2}.$$

# Problem 7

For two women $w$ and $w'$, we write $w <_m w'$ to denote that $w$ is worse than $w'$ in the preference list of man $m$. Given two stable matchings $f$ and $f'$ (you can easily construct an example in which there are multiple stable matchings), define a mapping $g = f \vee f'$ as follows:

- for each man $m$, assign him the more preferred partner:

$$g(m) = \begin{cases} f(m), & \text{if } f(m) \geq_m f'(m), \\ f'(m), & \text{if } f'(m) >_m f(m); \end{cases}$$

- for each woman $w$, assign her the less preferred partner:

$$g(w) = \begin{cases} f(w), & \text{if } f(w) \leq_w f'(w), \\ f'(w), & \text{if } f'(w) <_w f(w). \end{cases}$$

Show that if both $f$ and $f'$ are stable matchings, so is $g$.

*Note*: We can similarly define $f \wedge f'$. Then all stable matchings form a distributive lattice, an abstract yet very popular object studied in combinatorics.

**Answer:**

We first show that $g$ is a perfect matching. If two distinct men $x_1$ and $x_2$ were both assigned $y$, then, since $f$ and $f'$ are matchings, w.l.o.g. $f(x_1) = f'(x_2) = y$, $f'(x_1) <_{x_1} y$ and $f(x_2) <_{x_2} y$. If $x_1 <_y x_2$, then $(x_2, y)$ is unstable in $f$, since $f(x_2) <_{x_2} y$ and $f(y) = x_1 <_y x_2$; otherwise $(x_1, y)$ is unstable in $f'$, since $f'(x_1) <_{x_1} y$ and $f'(y) = x_2 <_y x_1$. Both contradict stability, so the men's rule is bijective.

Now let $x$ be the man assigned to a woman $y$, so $x \in \{f(y), f'(y)\}$; assume w.l.o.g. that $x_1 = f(y) \leq_y f'(y) = x_2$. If $x = x_2$, then $g(x_2) = y$ and $f'(x_2) = y$ give $f(x_2) <_{x_2} y$, which together with $f(y) = x_1 \leq_y x_2$ makes $(x_2, y)$ unstable in $f$, a contradiction. Hence $x = x_1$: every woman receives her less preferred partner, exactly as prescribed, and the two rules agree.

Finally, suppose a man $x$ and a woman $y$ form an unstable pair in $g$, i.e. $g(x) <_x y$ and $g(y) <_y x$. Since $g(x)$ is the more preferred of $f(x)$ and $f'(x)$, we have $f(x) <_x y$ and $f'(x) <_x y$; since $g(y)$ is the less preferred of $f(y)$ and $f'(y)$, we have $f(y) <_y x$ or $f'(y) <_y x$. In the first case $(x, y)$ is unstable in $f$, since $f(x) <_x y$ and $f(y) <_y x$; in the second case it is unstable in $f'$, since $f'(x) <_x y$ and $f'(y) <_y x$. Both contradict stability.

Thus $g$ is a perfect matching with no unstable pair, i.e. a stable matching.

# AI Disclosure

I used an AI assistant on all problems in this homework, for grammar polishing, LaTeX typesetting, and checking the correctness of my arguments. All mathematical ideas and proofs are my own.